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GRADE A+**
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Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 4

Title: Cleaning and Preparing Customer Sales Data Using Power Query

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GitHub Link: <https://github.com/laishramritikumarsingh/BA-Assignment>

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Cleaning and Preparing Customer Sales Data Using Power Query

Problem Statement

A company has received customer sales data containing missing values, duplicate records, and inconsistent entries. The objective is to clean and prepare the dataset using Power Query before performing business analysis. The dataset used for this experiment is `dirty_cafe_sales.csv`. The raw file contains 10,000 transaction records and eight fields.

Dataset Overview

Field	Description / Expected Type
Transaction ID	Unique transaction identifier – Text
Item	Cafe product – Text
Quantity	Number of items purchased – Whole Number
Price Per Unit	Unit price – Decimal Number
Total Spent	Transaction value – Decimal Number
Payment Method	Payment type – Text
Location	Sales channel/location – Text
Transaction Date	Transaction date – Date

Methodology / Steps Followed

Step 1: Data Import

Imported the CSV file into Excel Power Query using Data → Get Data → From Text/CSV → Transform Data.

Step 2: Initial Data Inspection

Reviewed the dataset and identified missing values, ERROR entries, UNKNOWN entries and data-type issues.

Step 3: Handling Invalid Entries

Replaced invalid ERROR values in numeric/date fields with null and standardized invalid text entries.

Step 4: Handling Missing Values

Handled missing categorical values using a consistent label such as Not Provided/Not Specified. Numeric and date fields without a reliable replacement were retained as null.

Step 5: Text Cleaning

Applied Trim and Clean transformations to remove unnecessary spaces and unwanted characters.

Step 6: Data Type Conversion

Converted Quantity to Whole Number, Price Per Unit and Total Spent to Decimal Number, and Transaction Date to Date.

Step 7: Duplicate Removal

Used Remove Duplicates to eliminate repeated transaction records.

Step 8: Final Validation

Verified the cleaned fields and checked that the resulting table contained no visible data-type errors.

Raw Dataset Before Cleaning

The following screenshot shows the original dataset before transformation, including ERROR, UNKNOWN and blank entries.

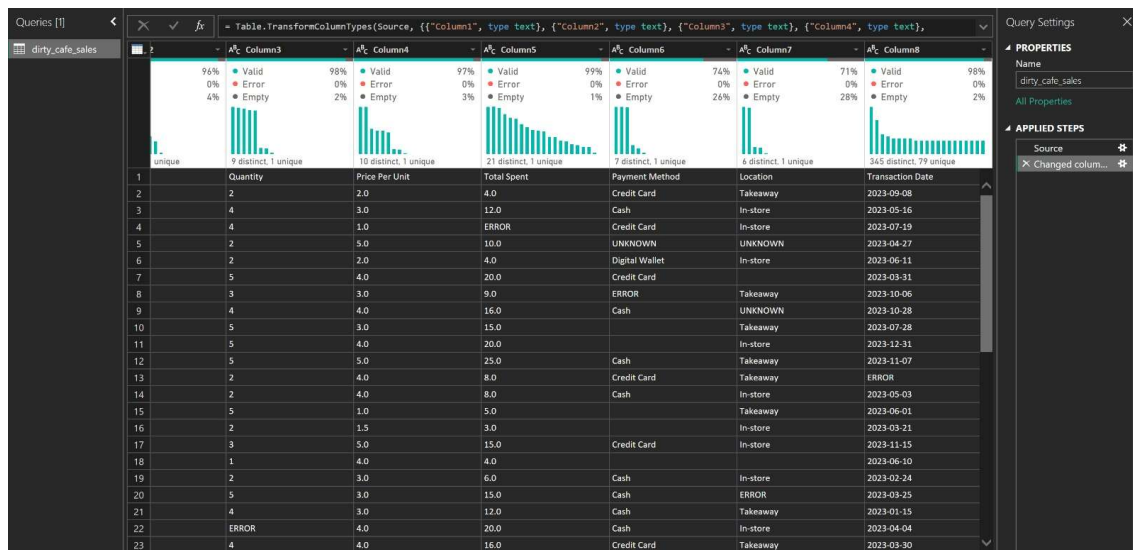


Figure 1: Raw Cafe Sales Dataset Before Cleaning

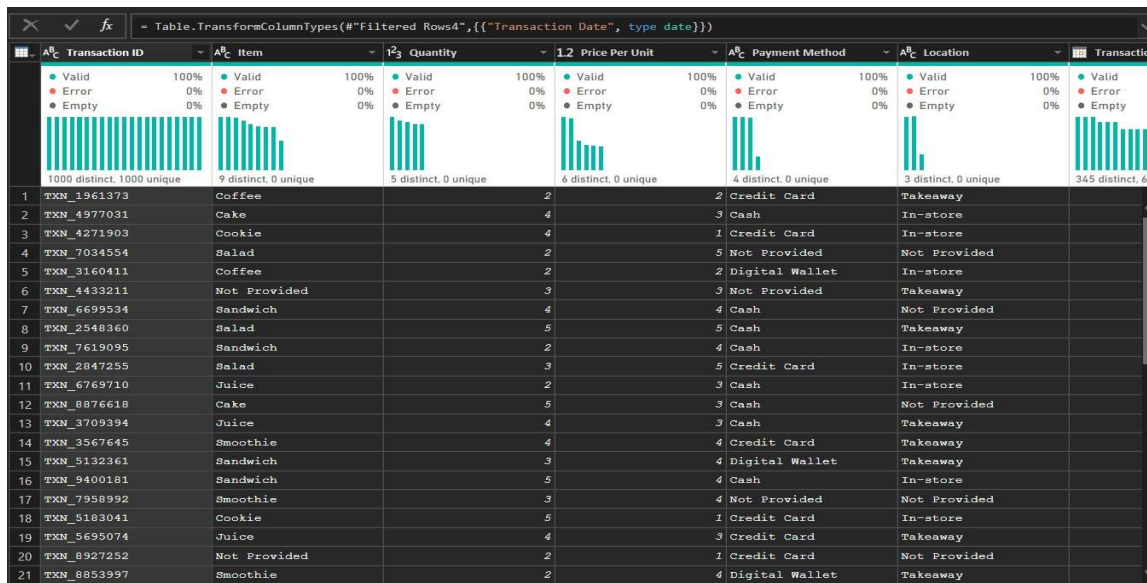


Figure 2: Cafe Sales Dataset After Cleaning

Output

The cleaned dataset was produced through a sequence of Power Query transformations. Invalid entries were standardized, text fields were cleaned, appropriate data types were assigned, and duplicate transaction records were removed.

Power Query Transformation Steps

The Applied Steps panel records the sequence of transformations used during the cleaning process.

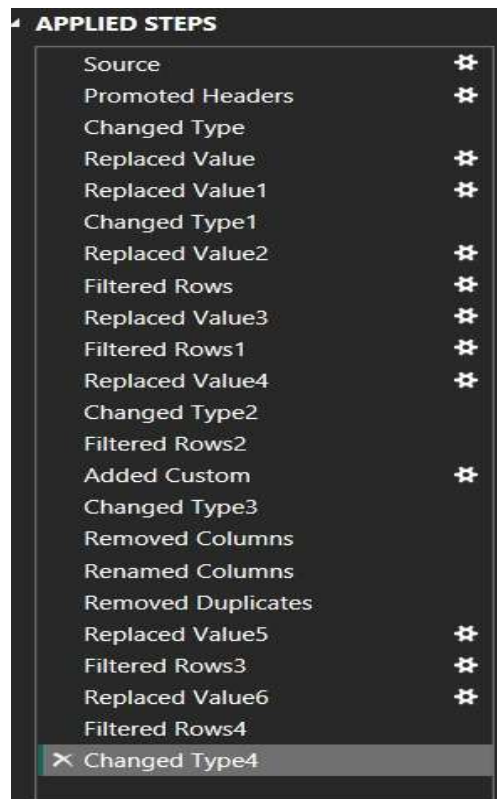


Figure 2: Power Query Applied Steps and Cleaned Data

Cleaning Summary

Data Quality Issue	Action Taken	Result
ERROR values	Replaced/handled before type conversion.	No visible errors
UNKNOWN values	Standardized to a consistent missing-value label.	Consistent categories
Missing categorical values	Handled using Not Provided/Not Specified.	Categorical fields standardized
Missing numeric/date values	Retained as null where no reliable value was available.	No false values introduced
Text inconsistencies	Applied Trim and Clean.	Cleaner text values
Duplicate records	Removed duplicate transaction records.	Duplicate check completed
Data types	Converted columns to appropriate types.	Analysis-ready structure

Final Data Quality Check

The final Power Query view showed 0% errors in the cleaned columns visible in the final transformation. Some numeric and date fields may remain empty because the source data did not contain a reliable value. Keeping such values as null avoids introducing inaccurate information.

Cleaning Workflow

Stage 1	Stage 2	Stage 3
Import Raw CSV	Inspect Data	Identify Quality Issues
↓	↓	↓
Handle ERROR / UNKNOWN	Handle Missing Values	Trim & Clean Text
↓	↓	↓
Correct Data Types	Remove Duplicates	Validate Dataset
↓	↓	↓
Final Clean Dataset	→	Ready for Analysis

Sample Cleaned Data

Transaction ID	Item	Quantity	Price/Unit	Payment Method	Location
TXN 1961373	Coffee	2	2.0	Credit Card	Takeaway
TXN 4977031	Cake	4	3.0	Cash	In-store
TXN 4271903	Cookie	4	1.0	Credit Card	In-store
TXN 7034554	Salad	2	5.0	Not Provided	Not Provided
TXN 3160411	Coffee	2	2.0	Digital Wallet	In-store

Conclusion

The cafe sales dataset was successfully cleaned and prepared using Power Query. The process addressed major data-quality issues including missing values, ERROR entries, UNKNOWN entries, text inconsistencies, data-type problems and duplicate records. The resulting dataset is structured for reliable business analysis and reporting.

Learning Outcomes

Through this experiment, I learned:

- How to import a CSV dataset into Power Query.
- How to identify and handle missing and invalid values.
- How to standardize inconsistent categorical entries.
- How to trim and clean text fields.
- How to assign appropriate data types to columns.
- How to check and remove duplicate transaction records.
- How Power Query prepares raw business data for analysis.
- How to validate a cleaned dataset before reporting.

Result

The final cleaned dataset can now be used for business analysis such as item-wise sales analysis, payment-method analysis, location/channel analysis and transaction trends.